A pointed map of pointed spaces yields a long exact sequence of homotopy groups

The Univalent Foundations Program

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Editorial note. The counted statement is the author's thm environment at source lines 1080-1093 and its proof environment opens at line 1100; lines 1095-1099 between them are a LaTeX comment, a vertical-space command and a page-break hint of the author, none of which belongs to either environment.

Editorial note. The retained counter values (chapter 8, section 4, theorem offset 5) make the counted theorem print as Theorem 8.4.6, matching the original book numbering read from the label-number dump of the pinned snapshot (newlabel thm:les = 8.4.6).

Editorial note. Three labels the proof cites lie outside the card and are declared in external_references as non-load-bearing navigation pointers: thm:path-truncation (book Theorem 7.3.12, defined at line 671 of the hlevels chapter's pinned file), thm:fiber-of-the-fiber (book Theorem 8.4.4, defined at line 1001 of this chapter's own pinned file) and thm:EckmannHilton (book Theorem 2.1.6, whose defining chapter is not part of the pinned corpus, so its pointer span is the label line of this chapter's label-number dump).

Editorial note. The retained preamble is the book's own macro library and layout adapter as stored for this chapter; it loads the xy package, which the retained statement needs for the diagram of the long exact sequence.

Editorial scope. One original result of the pinned HoTT Book source tree, counted once: the long exact sequence of homotopy groups associated with a pointed map (author label thm:les, book Theorem 8.4.6).

Allowed standard prerequisites (editorial).

- Fiber sequence of a pointed map. For a pointed map f : X \to Y the fiber F \defeq \hfib f {y_0} sits in a fiber sequence in which the next maps are \proj1 and the loop of f.. It is the input of the theorem (source lines 1080-1081).
- Exactness of a sequence of pointed sets. A sequence of pointed sets is exact at the middle term when the image of one map is contained in the kernel of the next and conversely.. It is what the proof establishes for the 0-truncations (source lines 1102-1108).
- Homotopy groups as truncated loop spaces. \pi_k is the 0-truncation of the k-fold loop space, so it is a group for k \ge 1 and abelian for k \ge 2 by the Eckmann-Hilton argument.. It supplies the group structure asserted in the statement (source lines 1119-1120).

Source reference. Original theorem 7.3.12: sources/hott-hlevels/pinned-original-source.tex, source lines 671–671. Book Theorem 7.3.12, that a truncated path type is the path type of the truncation, used at source line 1112 to lift a projected equality to a truncated path. Navigation pointer only: the counted proof is retained in full and the pointer span is the line of the label in the pinned file that carries that theorem's statement.

Source reference. Original theorem 8.4.4: sources/hott-homotopy/pinned-original-source.tex, source lines 1001–1001. Book Theorem 8.4.4, identifying the maps of the long exact sequence with the induced maps on homotopy groups, invoked at source line 1121. Navigation pointer only: it is an identification used in one step of an otherwise complete argument.

Source reference. Original theorem 2.1.6: sources/hott-homotopy/book-labelnumbers-first-edition.tex, source lines 1767–1767. Book Theorem 2.1.6, the Eckmann-Hilton argument, invoked at source line 1120 for the abelian structure of the higher homotopy groups. Navigation pointer only; its defining chapter is not part of the pinned corpus, so the pointer span is the label line of this chapter's label-number dump.

Theorem 8.4.6. *Let $f : X \to Y$ be a pointed map between pointed spaces with fiber $F :\equiv \mathsf{fib}_f(y_0)$. Then we have the following long exact sequence, which consists of groups except for the last three terms, and abelian groups except for the last six.*

$$\begin{array}{ccccc}
\vdots & & \vdots & & \vdots \\
\pi_k(F) & \longrightarrow & \pi_k(X) & \longrightarrow & \pi_k(Y) \\
\vdots & & \vdots & & \vdots \\
\pi_2(F) & \longrightarrow & \pi_2(X) & \longrightarrow & \pi_2(Y) \\
\pi_1(F) & \longrightarrow & \pi_1(X) & \longrightarrow & \pi_1(Y) \\
\pi_0(F) & \longrightarrow & \pi_0(X) & \longrightarrow & \pi_0(Y)
\end{array}$$

Proof. We begin by showing that the 0-truncation of a fiber sequence is an exact sequence of pointed sets. Thus, we need to show that for any adjacent pair of maps in a fiber sequence:

$$\mathsf{fib}_f(z_0) \xrightarrow{g} W \xrightarrow{f} Z$$

with $g :\equiv \mathsf{pr}_1$, the sequence

$$\|\mathsf{fib}_f(z_0)\|_0 \xrightarrow{\|g\|_0} \|W\|_0 \xrightarrow{\|f\|_0} \|Z\|_0$$

is exact, i.e. that $\mathsf{im}(\|g\|_0) \subseteq \ker(\|f\|_0)$ and $\ker(\|f\|_0) \subseteq \mathsf{im}(\|g\|_0)$.

The first inclusion is equivalent to $\|f\|_0 \circ \|g\|_0 = 0$, which holds by functoriality of $\|-\|_0$ and the fact that $f \circ g = 0$. For the second, we assume $w' : \|W\|_0$ and $p' : \|f\|_0(w') = |z_0|_0$ and show there merely exists $t : \mathsf{fib}_f(z_0)$ such that $|g(t)|_0 = w'$. Since our goal is a mere proposition, we can assume that w' is of the form $|w|_0$ for some $w : W$. Now by original theorem 7.3.12, $p' : |f(w)|_0 = |z_0|_0$ yields $p'' : \|f(w) = z_0\|_{-1}$, so by a further truncation induction we may assume some $p : f(w) = z_0$. But now we have $|(w, p)|_0 : \|\mathsf{fib}_f(z_0)\|_0$ whose image under $\|g\|_0$ is $|w|_0 \equiv w'$, as desired.

Thus, applying $\|-\|_0$ to the fiber sequence of f, we obtain a long exact sequence involving the pointed sets $\pi_k(F)$, $\pi_k(X)$, and $\pi_k(Y)$ in the desired order. And of course, π_k is a group for $k \geq 1$, being the 0-truncation of a loop space, and an abelian group for $k \geq 2$ by the Eckmann–Hilton argument (original theorem 2.1.6). Moreover, original theorem 8.4.4 allows us to identify the maps $\pi_k(F) \to \pi_k(X)$ and $\pi_k(X) \to \pi_k(Y)$ in this exact sequence as $(-1)^k \pi_k(i)$ and $(-1)^k \pi_k(f)$ respectively.

More generally, every map in this long exact sequence except the last three is of the form $\|\Omega h\|_0$ or $\|-\Omega h\|_0$ for some h. In the former case it is a group homomorphism, while in the latter case it is a homomorphism if the groups are abelian; otherwise it is an "anti-homomorphism". However, the kernel and image of a group homomorphism are unchanged when we replace it by its negative, and hence so is the exactness of any sequence involving it. Thus, we can modify our long exact sequence to obtain one involving $\pi_k(i)$ and $\pi_k(f)$ directly and in which all the maps are group homomorphisms (except the last three). □